Changes in composition and diversity of soil fungal communities along an age gradient of old-growth boreal forests in northwestern Quebec, Canada

Jonathan Cazabonne[1], Mélanie Roy[2,3], Irene Calderon-Sanou[4], Camille Revertegat[5], Clément Lionnel[6], Mickaël Hedde[7], Ludovic Gelly[6], Daniele Charron[1], Yves Bergeron[1], David Paré[8], Wilfried Thuiller[6], Maxence Martin[1]

[1]Université du Québec en Abitibi-Témiscamingue, [2]University Paul-Sabatier Toulouse III, [3]IRL IFAECI Instituto Franco-Argentino para el Estudio del Clima y sus Impactos, [4]IUMR EcoFoG, [5]IURFM Ecologie des Forêts Méditerranéennes, [6]IUMR LECA, [7]IUMR Eco&Sols, [8]Centre de foresterie des Laurentides

Context

Old-growth forests in eastern Canada are recognized as **unique reservoirs of biodiversity** and **hotspots of fungal diversity**, particularly for **ectomycorrhizal fungi**. Old-growth forests are increasingly recognized as highly heterogeneous and structurally diverse ecosystems, yet little is still known about how **soil fungal communities** vary among different types of old-growth forests.

Objectives

Our objective was to explore how **patterns of composition** and **diversity** in soil fungal communities vary among **old-growth boreal forest type**, and to test the **effect of time since the last fire** (TSF hereafter) on these patterns. The **dating of ancient fires** from 1944 to 1760 years ago found in the **Duparquet Lake Teaching and Research Forest** offers a unique opportunity to evaluate the dynamics of fungal communities in old-growth boreal forests.

Hypothesis

We hypothesized that **(1)** patterns of diversity, community composition, and relative abundances would be taxa- and guild-specific and would not necessarily follow linear trends along successional stages; **(2)** TSF would have significant effects on the diversity and community composition of soil fungal communities, notably through changes in soil physicochemical properties; and **(3)** regardless of successional patterns, the ectomycorrhizal fungi would remain the dominant guild along the post-fire chronosequence.

Material & Methods

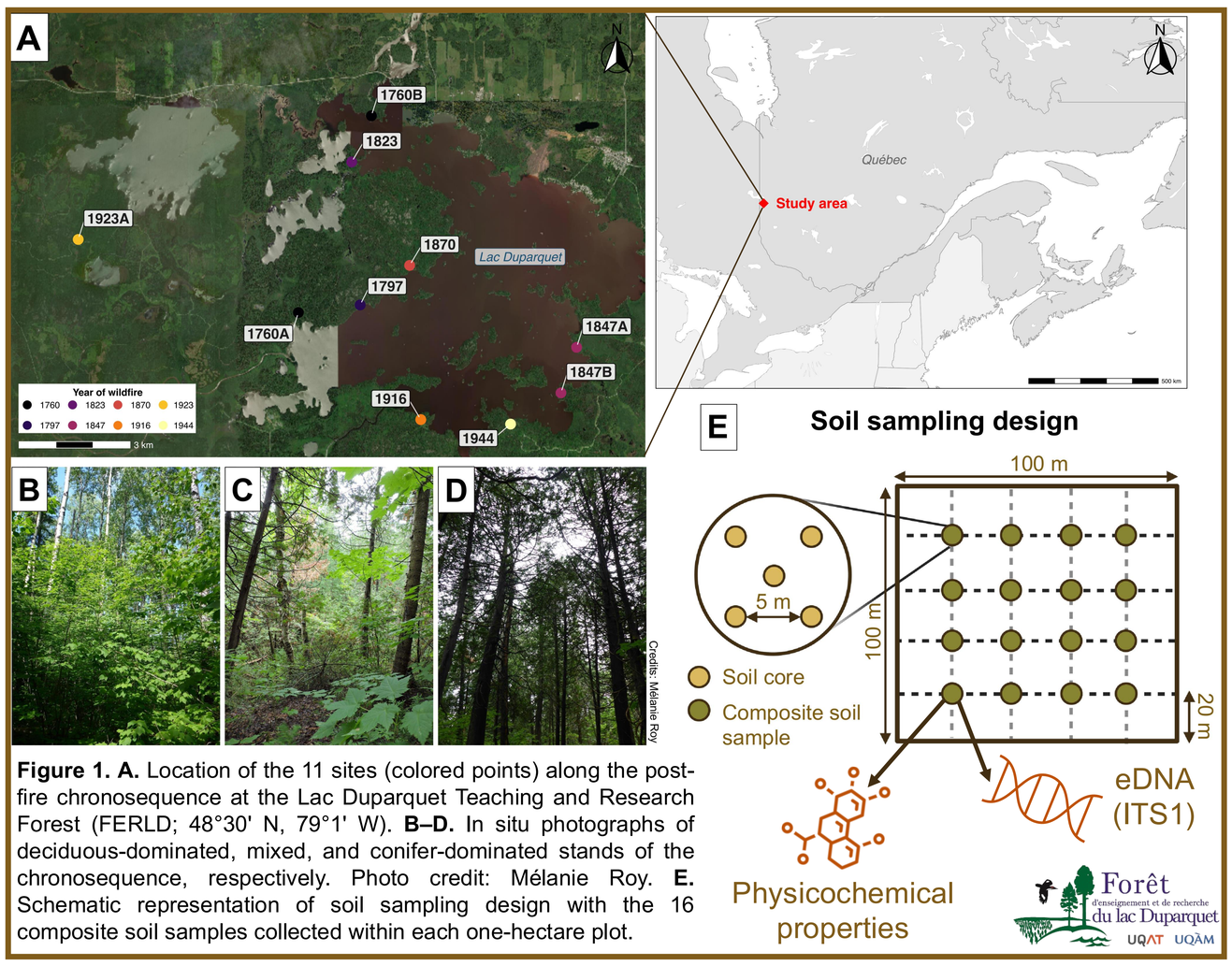


Figure 1. A. Location of the 11 sites (colored points) along the post-fire chronosequence at the Lac Duparquet Teaching and Research Forest (FERLD; 48°30' N, 79°1' W). **B–D.** In situ photographs of deciduous-dominated, mixed, and conifer-dominated stands of the chronosequence, respectively. Photo credit: Mélanie Roy. **E.** Schematic representation of soil sampling design with the 16 composite soil samples collected within each one-hectare plot.

Results

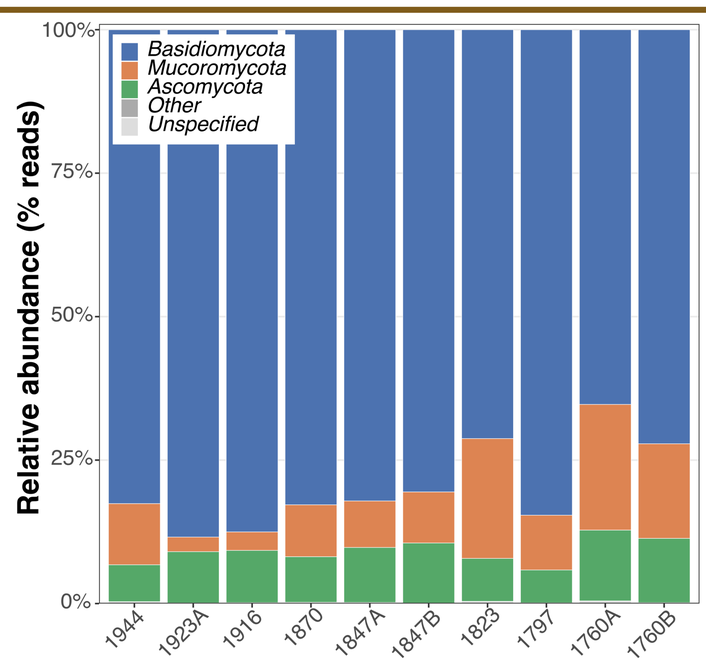


Figure 2. Stacked bar plots of the relative abundance of fungal phyla across the post-fire chronosequence.

- **Basidiomycota**, followed by **Mucoromycota** and **Ascomycota**, were the three most abundant phyla.
- Relative abundance of **Archaeorhizomycetes** significantly increased with TSF, whereas the relative abundance of **ectomycorrhizal fungi** significantly decreased with TSF (DESeq2).
- **Mucoromycota** and **arbuscular mycorrhizal fungi** tended to be more abundant in older old-growth stands.

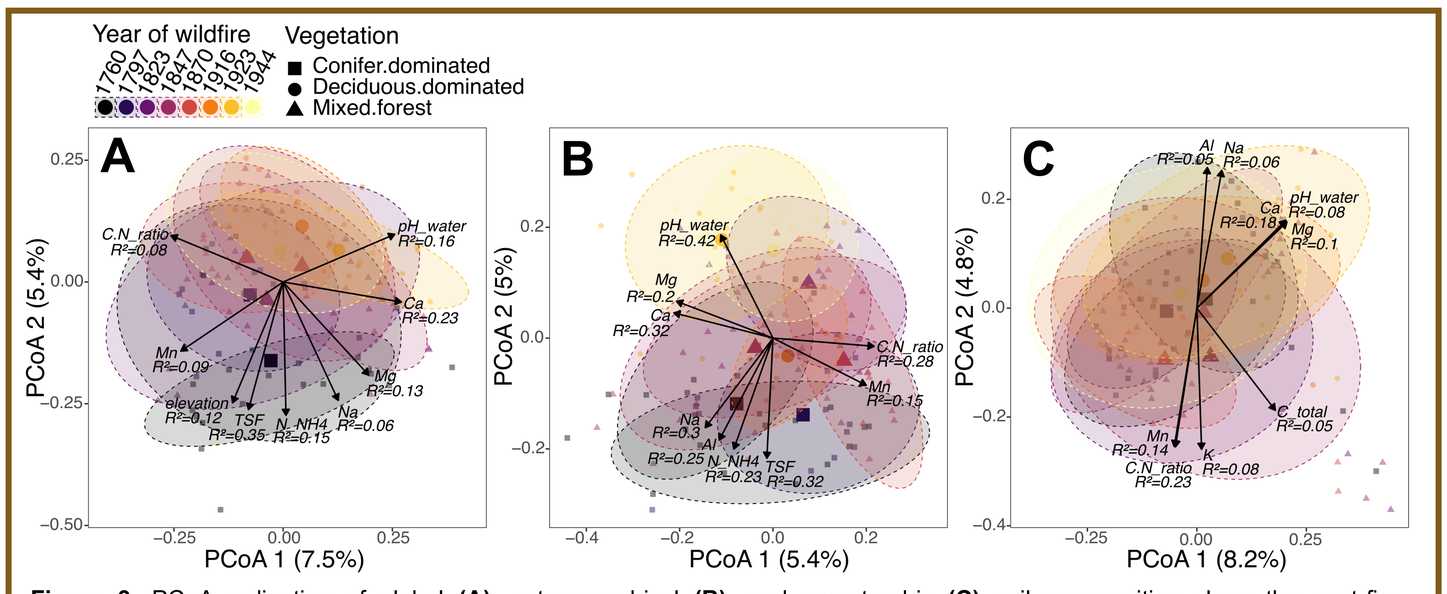


Figure 3. PCoA ordination of global **(A)**, ectomycorrhizal **(B)**, and saprotrophic **(C)** soil communities along the post-fire chronosequence, with physicochemical variables fitted using envfit (only variables with $p < 0.05$ are shown, along with R^2 values).

- Global, ectomycorrhizal, and saprotrophic **community composition differed significantly with TSF** (PERMANOVA test, $p < 0.01$).
- **Beta diversity** (distance to centroid), calculated using betadisper, increased significantly with TSF only for ectomycorrhizal fungi.
- The best-fitting **GAMs** for the global and ectomycorrhizal communities included only TSF as a significant factor influencing the three **Hill numbers (q0, q1, q2)**. For the saprotrophic community, the best-fitting GAMs for the three Hill numbers included as a significant factor only the first principal component (PC1) from a PCA of the **physicochemical variables**.

Conclusion & perspectives

Soil fungal communities are **highly diverse among old-growth forests** and **change across successional stages** alongside the vegetation shifts associated with TSF. However, the relative importance of TSF and soil physicochemical properties in structuring these communities is **taxon-dependent**, raising questions about the **environmental factors driving local-scale heterogeneity** in soil fungal communities within old-growth forests. This type of knowledge will be important for **sustainably integrating funga into forest management** by recognizing the diversity and complexity of soil fungal communities within old-growth forests and ultimately **developing tools to monitor** these overlooked yet essential components of forest biodiversity.

Thanks to our funding partners

anr
Fonds de recherche Québec
CRSNG NSERC
Ressources naturelles et Forêts Québec
CHANTIERS CHIBOUGAMAU



Scan me to learn more about my research!

jonathan.cazabonne@uqat.ca